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Research about Gravitational Wave Scattering

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ABSTRACT

Scattering in the background of a curved spacetime of a black hole has fundamental research significance. Since the mid-20th century, detailed studies have been conducted on non-time-dependent scattering. However, our research on different types of waves is limited, and a key obstacle is the lack of convergence in the summation of partial waves when calculating scattering amplitudes. Our work aims to construct a framework to address the aforementioned divergence problem—by abandoning the plane wave incidence assumption in traditional partial wave methods, considering the joint influence of long-range forces from the wave source and scattering source, analyzing the propagation process from the wave source to the scattering source and then to infinity to solve the problem of the lack of convergence in scattering amplitudes. Additionally, in cases where analytical solutions to differential equations are lacking, we aim to extract the necessary information about scattering and validate the convergence or divergence of our method through numerical methods. Simultaneously, the method can be extended to apply to electromagnetic waves and gravitational waves.

1. INTRODUCTION

Scattering is a fundamental manifestation of interactions between matter. Whether it's the exploration of new physics in high-energy physics colliders or the formation of the azure sky, scattering plays an indispensable role. In the cosmos, electromagnetic waves and gravitational waves permeate the universe and constantly undergo scattering with massive celestial bodies. Many phenomena, including gravitational lensing, are manifestations of scattering. If we aim to further understand the intrinsic information within the universe, studying the significance of scattering, including electromagnetic waves and gravitational waves, is crucial.

The fundamental method for studying scattering problems is the Partial Wave Method introduced in quantum mechanics. Its main idea is to separate the scattering process into two parts: one part is the plane wave propagating from infinity to the scattering source, and the other part is the scattering wave generated at the scattering source. We decompose the above two parts at the scattering source using them as a basis when considering some short-range potentials, resulting in an infinite series summation, where each term is called a partial wave. However, when dealing with long-range potential situations using the Partial Wave Method, convergence issues arise in the summation of partial waves. This fundamentally stems from the invalidity of the plane wave assumption under long-range potential conditions. Therefore, it is necessary to seek new physical scenarios and corresponding theoretical explanations.

Our theoretical framework departs from the plane wave assumption in the Partial Wave Method by considering the potential distribution resulting from the combined effects of the wave source and the scattering by special functions. We take into account the propagation process from the wave source to the scattering medium and towards infinity. To simplify the problem, we divide the entire system into a local region (where the short-range potential of the black hole dominates and relativistic effects are significant) and an approximation region (where the potential is approximated by classical Newtonian potential). In the approximation region, we consider general Newtonian potentials. According to the Schrödinger equation, we ultimately obtain elliptic functions that determine the behavior of the wave function, and this differential equation constitutes an eigenvalue problem. However, our study of the elliptic equation itself is not sufficient. Difficulties in determining boundary conditions prevent us from directly obtaining analytical solutions to the equation. Therefore, obtaining information about the scattering waves without solving the equation is our main focus.

For the local region, we consider the most general case of scattering by Kerr black holes (rotating black holes). At the scattering source, the gravitational influence of a single Kerr black hole predominates. There are already well-established methods for calculating the behavior of the wave function near Kerr black holes, particularly represented

by the Teukolsky equation theory , which fully describes the behavior near Kerr black holes. Of particular interest to us is the phase change generated by the scattering of waves near black holes. To simplify the problem, we can directly represent the behavior before and after scattering using phase shifts, establishing a correspondence with the expression determining the behavior of the wave function in the approximation region. However, calculating phase shifts still poses theoretical challenges. Here, we choose to approximate the phase shift calculation using the WKB method .

In summary, our theoretical framework provides a promising solution to the convergence issues encountered in scattering problems. Moreover, the theory is sufficiently foundational to be extended to a variety of applications, including interactions between electromagnetic waves and massive celestial bodies such as gravitational lensing, as well as interactions between gravitational waves and compact objects. Our framework can be applied to calculate these interactions and provide fundamental theoretical support for observational astronomy.

2. THEORETICAL PART

Due to the long-distance effects of gravitational interactions, it is not sufficient to simply use plane waves as incident waves. Instead, one needs to consider the gravitational potentials of both the source and the scattering center. We first assume that two celestial bodies, A and B, are sufficiently far apart, allowing us to treat different regions differently. Beyond a certain long dashed line, we only need to consider the long-range potential of the wave source and scattering center, namely the Newtonian potential part, which we refer to as the long-range region. Within this region, the dominant influence is the long-range potential of both celestial bodies. Within the long dashed line but outside the short dashed line, due to the significant distance between the two bodies, only the long-range potential of one celestial body dominates. This region is referred to as the approximation region. Within the short dashed line, close to the central body, we need to consider the short-range potential effects brought about by general relativity. This region is named the local region.

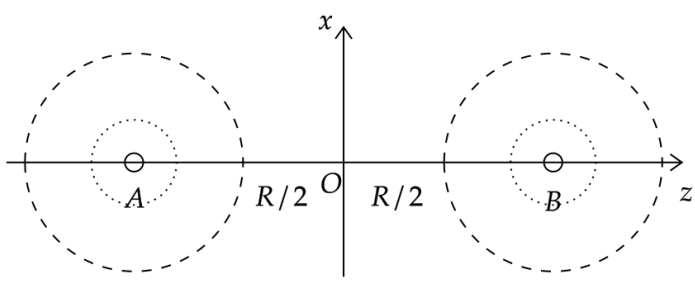

Figure 1. Approximation Region and Local Region Illustration

2.1. *Long Range Eqaution*

First, let's consider the equations in the approximation region. Considering the system of the wave source-scattering center illustrated in Figure 1 below, where point A is the wave source and point B is the scattering center. For simplicity, we begin by considering the scalar wave scattering in flat spacetime governed by the Schrödinger equation. The potential is given by the Coulomb potential with strengths N_A and N_B.

$$\left(-\frac{1}{2M}\nabla^2 - \frac{N_A}{r} - \frac{N_B}{r}\right)\psi = \epsilon\psi \tag{1}$$

Where ϵ represents the energy of the scattering state. We perform a transformation of the above equations into ellipsoidal coordinates, which are defined as follows:

$$\mu = \frac{r_A + r_B}{R},\ \lambda = \frac{r_A - r_B}{R},\ \phi \tag{2}$$

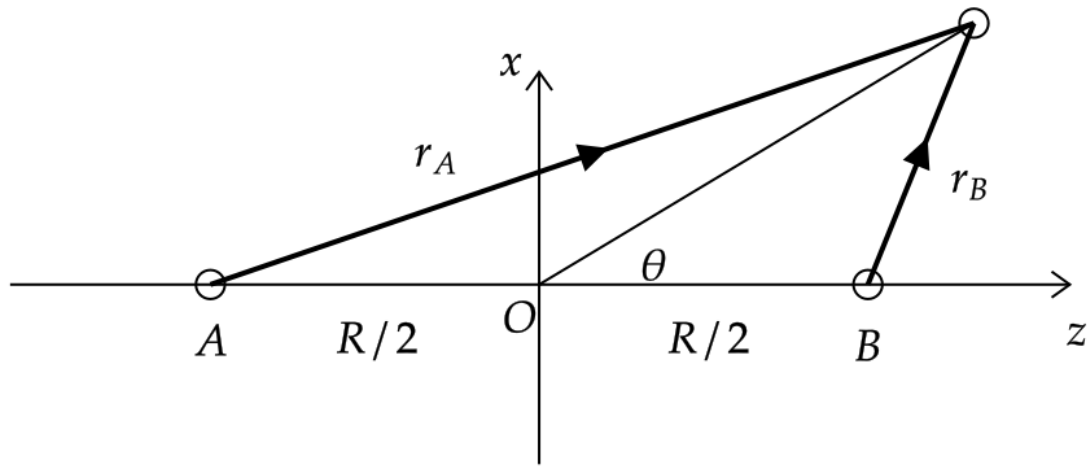

Figure 2. The system constructed by the wave source and the scattering center.

The result of the action of the Laplace-Beltrami operator on a scalar field is given by:

$$\nabla^2\psi = \frac{1}{\sigma^2\left(\mu^2-\lambda^2\right)}\left\{\frac{\partial}{\partial\mu}\left[\left(\mu^2-1\right)\frac{\partial\psi}{\partial\mu}\right]+\frac{\partial}{\partial\lambda}\left[\left(1-\lambda^2\right)\frac{\partial\psi}{\partial\lambda}\right]\right\} \tag{3}$$

$$+\frac{1}{\sigma^2\left(\mu^2-1\right)\left(1-\lambda^2\right)}\frac{\partial^2\psi}{\partial\varphi^2} \tag{4}$$

Given that $\sigma = \frac{R}{2}$, we can perform a separation of variables:

$$\psi = \sum_m a_m F_m(\mu) G_m(\lambda) e^{im\varphi} \tag{5}$$

This allows us to derive corresponding differential equations. This set of equations is also known as the ellipsoidal equations:

$$\left[\frac{d}{d\mu}\left(1-\mu^2\right)\frac{d}{d\mu}+\frac{m^2}{1-\mu^2}-\rho_m+2M\sigma\left(N_A+N_B\right)\mu+2M\sigma^2\epsilon\mu^2\right]F_m(\mu)=0 \tag{6}$$

$$\left[\frac{d}{d\lambda}\left(1-\lambda^2\right)\frac{d}{d\lambda}-\frac{m^2}{1-\lambda^2}-\rho_m-2M\sigma\left(N_A-N_B\right)\lambda-2M\sigma^2\epsilon\lambda^2\right]G_m(\lambda)=0 \tag{7}$$

Here, ρ represents the eigenvalue of this differential equation. Since we are considering the scattering state, the eigenvalues are continuous. Therefore, the solution to the wave function should be written in integral form:

$$\psi = \sum_m \int d\rho a_m(\rho) F_m(\rho,\mu) G_m(\rho,\lambda) e^{im\phi} \tag{8}$$

2.2. *Analysis of Spheroidal Equation*

Regarding the ellipsoidal equations, our understanding of their solutions is limited. This includes difficulties in determining the boundary conditions for the entire system in ellipsoidal coordinates. Therefore, we attempt to obtain the desired scattering information without directly solving the equations.

First, let's consider the behavior of waves in the approximation region near the wave source A, which is governed by the equation:

$$\frac{d^2}{dx^2}H_l^{A\pm}(x)+\left[1-\frac{l(l+1)}{x^2}-\frac{2\lambda_A}{x}\right]H_l^{A\pm}(x)=0 \tag{9}$$

Here, $H_l^{A\pm}(x)$ represents the confluent hypergeometric function. Comparing with Equation 1, we can determine the parameter relations: $x=\frac{\omega r_A}{\sigma}$ and $\lambda_A=-\sqrt{\frac{M}{2\epsilon}}N_A$. For the confluent hypergeometric function, it has an asymptotic form as $x\to\infty$:

$$H_l^{A\pm}(x)\to e^{\pm i[x-\lambda_A\log(2x)-\frac{l}{2}\pi+\sigma_l(\lambda_A)]} \tag{10}$$

Through this asymptotic form, we realize that for practical scattering problems, we cannot obtain plane wave solutions; instead, their asymptotic propagation forms are used. Additionally, we can establish a connection between the approximation region near A in the long-range region: when approaching A, the effect of B should be ignored, and instead, the wave function should propagate near A in the form of A's Coulomb wave. Therefore, we have the relationship:

$$\frac{1}{\sigma(\mu+\lambda)}\mathcal{F}_0^A(\omega(\mu+\lambda))=\int d\rho a_{l0}(\rho)F_0^{(\lambda_A+\lambda_B)}(\rho,1)G_0^{(\lambda_A-\lambda_B)}(\rho,\lambda) \tag{11}$$

Where,

$$\mathcal{F}_0^A(x)=\frac{e^{-\pi\lambda_A/2}\left|\Gamma\left(l+1+i\lambda_A\right)\right|}{2(2l+1)!}(-i)^{l+1}M_{i\lambda_A,l+\frac{1}{2}}(2ix)=\frac{1}{2i}\left[H_l^{A+}(x)-H_l^{A-}(x)\right] \tag{12}$$

For Equation 11, we have already indicated the dependence of F_m and G_m on λ_A and λ_B with superscripts, which can be easily derived from Equation 7. As waves propagate from the approximation region near point A to the long-range region, when $\lambda\sim-1$ and $1\ll\omega(\mu-1)\ll 2\omega$, we should have the corresponding relation:

$$\frac{1}{\sigma(\mu+\lambda)}[2\omega(\mu+\lambda)]^{-i\lambda_A}e^{i[\omega(\mu+\lambda)+\sigma_0(\lambda_A)]}=\int d\rho a_{l0}(\rho)F_0^{(\lambda_A+\lambda_B)}(\rho,1)G_0^{(\lambda_A-\lambda_B)}(\rho,\lambda) \tag{13}$$

For the above equation, we cannot directly obtain its analytical solution. However, based on Leaver's series solution method, we can obtain a series solution expanded near point A. For $m=0$, we have:

$$G_0(\rho,\lambda)=e^{i\omega(\lambda+1)}\sum_{n=0}^{+\infty}c_n^\lambda(\rho)(\lambda+1)^n \tag{14}$$

$$\begin{aligned}&\alpha_0^\theta c_1^\theta && +\beta_0^\theta c_0^\theta+ && =0,\\ &\alpha_n^\theta c_{n+1}^\theta && +\beta_n^\theta c_n^\theta+\gamma_n^\theta c_{n-1}^\theta && =0, n=1,2,\cdots\end{aligned} \tag{15}$$

Where the coefficients are given by:

$$\begin{aligned}&\alpha_n^\lambda=-2n^2-\left(B_2+2\right)n-B_2,\\ &\beta_n^\lambda=n^2+\left(B_2-4i\omega-1\right)n-a-i\omega B_2+\rho+m(m+1),\\ &\gamma_n^\lambda=2i\omega n+i\omega\left(B_2-2\right)+a,\end{aligned} \tag{16}$$

And the constants are:

$$
\begin{aligned}
a &= 2M\sigma\left(N_A - N_B\right) = -2\omega\left(\lambda_A - \lambda_B\right), \\
B_2 &= 2(m+1).
\end{aligned}
\tag{17}
$$

We can obtain the series coefficients, assuming the 0th order is $c_0^\lambda = 1$:

$$
\begin{aligned}
c_1^\lambda &= \frac{\rho}{2} - \frac{1}{2}(a + 2i\omega) \\
c_2^\lambda &= \frac{1}{16}\rho^2 + \frac{1}{8}(1 - a - 4i\omega)\rho + \frac{1}{16}\left(a^2 + 8ia\omega - 12\omega^2\right) \\
c_3^\lambda &= \frac{1}{288}\rho^3 + \frac{1}{288}(8 - 3a - 18i\omega)\rho^2 + \frac{1}{288}\left(3a^2 + 36ia\omega - 6a - 92\omega^2 - 36i\omega + 12\right)\rho \\
&\quad - \frac{1}{288}\left(a^3 + 18ia^2\omega + 2a^2 - 92a\omega^2 - 120i\omega^3 + 8\omega^2\right) \\
&\quad \cdots
\end{aligned}
\tag{18}
$$

Then we have the expression:

$$
G_0^{\lambda_A - \lambda_B}(\rho, \lambda) = e^{i\omega(\lambda+1)} \sum_{n=0}^{+\infty} c_n^\lambda(\rho)(1+\lambda)^n \tag{19}
$$

Although the series coefficients have been normalized, to avoid confusion in notation, we use $G_0(\rho, -1)$ as the normalization coefficient:

$$
G_0^{\lambda_A - \lambda_B}(\rho, \lambda) = G_0^{\lambda_A - \lambda_B}(\rho, -1) e^{i\omega(\lambda+1)} \sum_{n=0}^{+\infty} c_n^\lambda(\rho)(1+\lambda)^n \tag{20}
$$

Observing the coefficients of each order, we can easily see that ρ exists in a linearly superimposed form of power series. Therefore, to simplify the problem, we define the "n-th order moment" Z_n^A as:

$$
Z_n^A \equiv \int d\rho a_0(\rho) F_0^{\lambda_A + \lambda_B}(\rho, \mu) G_0^{\lambda_A - \lambda_B}(\rho, -1)\rho^n \tag{21}
$$

Then the right-hand side of equation 11 can be rewritten as:

$$
\begin{aligned}
e^{i\omega(\lambda+1)} \Big\{ & Z_0^A + \frac{1}{2}(\lambda+1)\left[-(a+2i\omega)Z_0^A + Z_1^A\right] \\
& + \frac{1}{16}(\lambda+1)^2\left[\left(a^2 + 8ia\omega - 12\omega^2\right)Z_0^A + 2(1 - a - 4i\omega)Z_1^A + Z_2^A\right] \\
& + \frac{1}{288}(\lambda+1)^3\left[-\left(a^3 + 18ia^2\omega + 2a^2 - 92a\omega^2 - 120i\omega^3 + 8\omega^2\right)Z_0^A\right. \\
& \left. + \left(3a^2 + 36ia\omega - 6a - 92\omega^2 - 36i\omega + 12\right)Z_1^A + (8 - 3a - 18i\omega)Z_2^A + Z_3^A\right] \\
& + \mathcal{O}\left((\lambda+1)^4\right)\Big\}
\end{aligned}
\tag{22}
$$

In fact, we are actually expanding the wave function around point A using the "n-th order moment" Z_n^A as a basis. If we want to solve for Z_n^A, it is not difficult to obtain. Firstly, when the above expansion should be equal to the Coulomb wave solution at point A, it should correspond to equation 13, and the above expression is a power series summation about $(1+\lambda)$. Then we can perform a Taylor expansion of equation 13 about $(1+\lambda)$, which will give us the expressions for each order of moments. We can write:

$$Z_0^A = \int d\rho a_0(\rho) F_0^{\lambda_A+\lambda_B}(\rho,\mu) G_0^{\lambda_A-\lambda_B}(\rho,-1) = \frac{[2\omega(\mu-1)^{-i\lambda_A}]}{\sigma(\mu-1)} e^{i[\omega(\mu-1)+\sigma_0(\lambda_A)]} \tag{23}$$

$$Z_1^A = \int d\rho a_0(\rho) F_0^{\lambda_A+\lambda_B}(\rho,\mu) G_0^{\lambda_A-\lambda_B}(\rho,-1)\rho \tag{24}$$

$$= \frac{[2\omega(\mu-1)^{-i\lambda_A}]}{\sigma(\mu-1)^2} e^{i[\omega(\mu-1)+\sigma_0(\lambda_A)]}[a(\mu-1) - 2i\lambda_A + 2i\omega(\mu-1) - 2] \tag{25}$$

It is evident that applying a similar approach near point B yields a similar scenario. However, the situations at points A and B are not directly comparable. This is because point A, being the source of the wave, only contains information about the outgoing waves, whereas for point B, the scattering source, it should include information about both incoming and reflected waves. Therefore, the expression at point A should be a degenerate form of the expression at point B. Thus, we need to find the relationship between the outgoing waves at point A and the wave function at point B. Firstly, we expand the wave near point B ($\lambda \sim 1$) in a similar manner:

$$G_0(\rho,\lambda) = G_0(\rho,1) e^{i\omega(1-\lambda)} \sum_{n=0}^{+\infty} d_n^\lambda(\rho)(1-\lambda)^n \tag{26}$$

Similarly, we have the same type of "n-th order moments" Z_n^B:

$$Z_n^B \equiv \int d\rho a_0(\rho) F_0^{\lambda_A+\lambda_B}(\rho,\mu) G_0^{\lambda_A-\lambda_B}(\rho,1)\rho^n \tag{27}$$

Therefore, we also have a similar expansion based on Z_n^B. However, it is evident that the relationship between Z_n^B and Z_n^A is not straightforward, particularly given the complexity of the integrals involved. Yet, noticing that $G_0^{\lambda_A-\lambda_B}(\rho,-1)$ is a function of ρ and $\lambda_A - \lambda_B$, we may assume that $G_0^{\lambda_A-\lambda_B}(\rho,-1)$ is related to $G_0^{\lambda_A-\lambda_B}(\rho,1)$.

$$G_0^{\lambda_A-\lambda_B}(\rho,1) = \xi(\rho) G_0^{\lambda_A-\lambda_B}(\rho,-1) \tag{28}$$

where, $\xi(\rho)$ is an undetermined function. However, it is certain that $\xi(\rho)$, being nontrivial, will provide information about the simultaneous presence of outgoing and incoming waves at point B. We have not yet obtained the method for computing $\xi(\rho)$, but what we expect is that in the far-field region, the expansion based on Z_n^B needs to match the expression in the vicinity of point B. This expression is:

$$\sum_l \frac{1}{\sigma(\mu-\lambda)} \left\{ u_l [2\omega(\mu-\lambda)]^{-i\lambda_B} e^{i[\omega(\mu-\lambda)+\sigma_l(\lambda_B)]} + v_l [2\omega(\mu-\lambda)]^{i\lambda_B} e^{-i[\omega(\mu-\lambda+\sigma_l(\lambda_B))]} \right\} P_l^0\left(\frac{\lambda\mu-1}{\mu-\lambda}\right) \tag{29}$$

Expanding around $\lambda \sim 1$ yields:

$$\frac{u_0}{\sigma(\mu-1)} [2\omega(\mu-1)]^{-i\lambda_B} e^{i[\omega(\mu-1)+\sigma_0(\lambda_B)]} + \frac{v_0}{\sigma(\mu-1)} [2\omega(\mu-1)]^{-i\lambda_B} e^{-i[\omega(\mu-1)+\sigma_0(\lambda_B)]} \tag{30}$$

2.3. *Local Field Equation*

For the local field analysis, we already have a relatively complete theory to explain it. In the case of the local region around point B, we first assume it to be a Kerr black hole (rotating black hole). We start with the simplest scalar field, and based on the work by Teukolsky , we utilize the Newman-Penrose formalism (abbreviated as N-P formalism) to address our problem. First, we write down the Kerr metric:

$$\begin{aligned} \mathrm{d}s^2 = &-\left(1 - \frac{2Mr}{\Sigma}\right)\mathrm{d}t^2 - \frac{4aMr\sin^2\theta}{\Sigma}\mathrm{d}t\ \mathrm{d}\phi + \frac{\Sigma}{\Delta}\mathrm{d}r^2 \\ &+ \Sigma \mathrm{d}\theta^2 + \left(\Delta + \frac{2Mr\left(r^2+a^2\right)}{\Sigma}\right)\sin^2\theta \mathrm{d}\phi^2 \end{aligned} \tag{31}$$

where

$$\begin{aligned}\Sigma &= r^2 + a^2\cos^2\theta \\ \Delta &= r^2 - 2Mr + a^2\end{aligned} \tag{32}$$

We construct a set of four frames that satisfy orthogonal relationships, which satisfy

$$l \cdot n = 1 \tag{33}$$

$$m \cdot m^* = -1 \tag{34}$$

Thus, we have the Newman-Penrose (N-P) formalism vector expressions

$$\begin{aligned}l^\mu &= [(r^2+a^2)/\Delta, 1, 0, a/\Delta] \\ n^\mu &= [r^2+a^2, -\Delta, 0, a] \\ m^\mu &= [ia\sin\theta, 0, 1, i/\sin\theta]/[2^{1/2}(r+ia\cos\theta)]\end{aligned} \tag{35}$$

We also define the following N-P formalism spin coefficient:

$$\begin{aligned}\rho &= -1/(r - ia\cos\theta) \\ \beta &= -\rho^*\cot\theta/(2\sqrt{2}) \\ \pi &= ia\rho^2\sin\theta/\sqrt{2} \\ \tau &= -ia\rho\rho^*\sin\theta/\sqrt{2} \\ \mu &= \rho^2\rho^*\Delta/2 \\ \gamma &= \mu + \rho\rho^*(r-M)/2 \\ \alpha &= \pi - \beta^*\end{aligned} \tag{36}$$

There are also corresponding N-P formalism differential operators:

$$\begin{aligned}D &= l^\mu\partial/\partial x^\mu \\ \Delta &= n^\mu\partial/\partial x^\mu \\ \delta &= m^\mu\partial/\partial x^\mu\end{aligned} \tag{37}$$

Substituting the Einstein field equations after variable substitution will result in a partial differential equation. Due to rotational symmetry, we can separate the variables. This yields a partial differential equation of the form:

$$\begin{aligned}&\left[\frac{\left(r^2+a^2\right)^2}{\Delta} - a^2\sin^2\theta\right]\frac{\partial^2\psi}{\partial t^2} + \frac{4Mar}{\Delta}\frac{\partial^2\psi}{\partial t\partial\varphi} + \left[\frac{a^2}{\Delta} - \frac{1}{\sin^2\theta}\right]\frac{\partial^2\psi}{\partial\varphi^2} \\ &-\Delta^{-s}\frac{\partial}{\partial r}\left(\Delta^{s+1}\frac{\partial\psi}{\partial r}\right) - \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) - 2s\left[\frac{a(r-M)}{\Delta} + \frac{i\cos\theta}{\sin^2\theta}\right]\frac{\partial\psi}{\partial\varphi} \\ &-2s\left[\frac{M\left(r^2-a^2\right)}{\Delta} - r - ia\cos\theta\right]\frac{\partial\psi}{\partial t} + \left(s^2\cot^2\theta - s\right)\psi = 4\pi\Sigma T.\end{aligned} \tag{38}$$

where

$$\psi = e^{i\omega t}e^{im\phi}S(\theta)R(r) \tag{39}$$

This leads to radial and angular differential equations.

$$\Delta^{-s}\frac{d}{dr}\left(\Delta^{s+1}\frac{dR}{dr}\right) + \left(\frac{K^2 - 2is(r-M)K}{\Delta} + 4is\omega r - \lambda\right)R = 0 \tag{40}$$

$$\begin{aligned}&\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dS}{d\theta}\right) + \\ &\left(a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta} - 2a\omega s\cos\theta - \frac{2ms\cos\theta}{\sin^2\theta} - s^2\cot^2\theta + s + A\right)S = 0,\end{aligned} \tag{41}$$

Here, $K \equiv (r^2 + a^2)\omega - am$, $\lambda \equiv A + a^2\omega^2 - 2am\omega$, and $\Delta = r^2 - 2Mr + a^2$.

The angular solutions are known as Spin-weighted Spheroidal Functions, which have been extensively studied in previous works. However, the radial equation, known as the Teukolsky equation, does not have a complete analytical solution. Nevertheless, some special forms of solutions have been obtained through research, such as confluent Heun functions.

Regarding the boundary conditions near the event horizon of a black hole, it has been found that:

$$\lim_{r \to r_+} R_{lm}(r) \sim (r - r_+)^{-i\sigma} \tag{42}$$

where $\sigma = \frac{2r_+(\omega - \omega_c)}{r_+ - r_-}$, $\omega_c = \frac{am}{2Mr_+}$, and $r_\pm = M \pm \sqrt{M^2 - a^2}$ (in units where $G = 1, c = 1$).

Since the Teukolsky equation does not have a complete analytical solution, numerical integration is typically employed to understand the behavior of the wave function in the local region. We perform these calculations using Mathematica code.

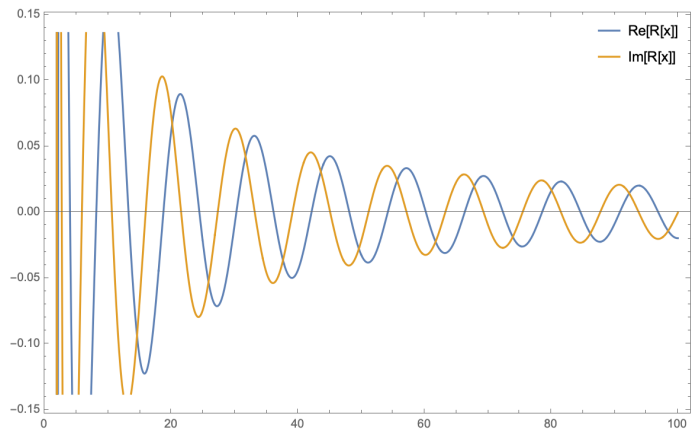

Figure 3. Radial Distribution for $a = 0.5, \omega - 0.5, l = 0, m = 0$

3. LOCAL FIELD SCATTERING PHASE SHIFT

In scattering processes, especially in astrophysical scenarios like gravitational wave scattering, we are often more concerned with the scattering phase shift and waveform information rather than the scattering cross-section. This is because for long-range potentials like gravitational or Coulomb potentials, the scattering cross-section tends to infinity, rendering it less meaningful for discussion. Here, we continue to consider the simplest case of scalar field scattering.

In the previous section, we obtained the distribution of wave functions near a Kerr black hole. It is worth noting that as $r \to \infty$, the general relativistic effects near the black hole, including the dragging effect due to black hole rotation, tend to zero, leaving only the Newtonian potential. This transition from the local to the far region implies that the asymptotic form of the Teukolsky equation at $r \to \infty$ must correspond to the confluent hypergeometric equation. In fact, after calculation, we obtain the asymptotic form of the Teukolsky equation as follows:

$$\lim_{r \to \infty} \left[\frac{(r^2 + a^2)^2}{\Delta} - a^2 \sin^2\theta \right] \frac{\partial^2 \psi}{\partial t^2} + \cdots \tag{43}$$

This asymptotic form indicates the behavior of the Teukolsky equation in the far region, where the gravitational effects become negligible and the equation reduces to a simpler form.

$$\frac{d^2}{dr^2}(rR) + \left[\omega^2 + \frac{4M\omega^2}{r} - \frac{\Lambda}{r^2} \right] rR = 0 \tag{44}$$

Where Λ is the eigenvalue of this equation, the expression is consistent with our conclusion.

Considering the phase shift, we can examine the process of scattering from infinity into the local region and then back to infinity. In this process, there will be a phase shift between the incident and scattered waves. Additionally,

due to the absorption of some waves by the black hole, the phase shift is complex and its magnitude is always less than 1. Moreover, the analytical form of the waves at infinity is represented by confluent hypergeometric functions or their asymptotic forms. We can express the expressions for the incident and scattered waves as follows, assuming the generated phase shift is:

$$\delta_l = \sigma_l^{LR} + \delta_l^{SR} \tag{45}$$

Here, σ_l^{LR} represents the phase shift caused solely by Coulomb wave scattering, which is known and appears in the asymptotic form of the confluent hypergeometric function, as shown in Equation 10. Its calculation is given by:

$$\sigma_l^{LR}(\lambda) \equiv \mathrm{Arg}(\Gamma(l+1+\lambda)) \tag{46}$$

However, the short-range potential part influenced by the black hole is unknown, and we need to isolate it. Considering the general form... (continues with the untranslated part)

$$u_L(x) \sim Ae^{2i\delta_L^{SR}} e^{i(x-\lambda\log(2x)+\sigma_L-\frac{1}{2}L\pi)} + Be^{-i(x-\lambda\log(2x)+\sigma_L-\frac{1}{2}L\pi)} \tag{47}$$

$$u_L(x) \sim B\left[\frac{A}{B}e^{2i\delta_L^{SR}} e^{i(x-\lambda\log(2x)+\sigma_L-\frac{1}{2}L\pi)} + e^{-i(x-\lambda\log(2x)+\sigma_L-\frac{1}{2}L\pi)}\right] \tag{48}$$

$$u_L'(x) \sim B\left[\frac{A}{B}e^{2i\delta_L^{SR}} e^{i(x-\lambda\log(2x)+\sigma_L-\frac{1}{2}L\pi)} i(1-\frac{\lambda}{x}) + e^{-i(x-\lambda\log(2x)+\sigma_L-\frac{1}{2}L\pi)}(1-\frac{\lambda}{x})(-i)\right] \tag{49}$$

Then, we can obtain:

$$u_L - \frac{iu_L'}{1-\frac{\lambda}{x}} = 2\hat{B}\frac{A}{B}e^{2i\delta^{SR}} e^{i(x-\lambda\log(2x)+\sigma_L-\frac{1}{2}L\pi)} \tag{50}$$

$$u_L + \frac{iu_L'}{1-\frac{\lambda}{x}} = 2\hat{B}e^{-i(x-\lambda\log(2x)+\sigma_L-\frac{1}{2}L\pi)} \tag{51}$$

Eliminating the amplitude information, we can obtain the phase shift information:

$$\frac{A}{B}e^{2i\delta^{SR}} = \frac{u_L - \frac{iu_L'}{1-\frac{\lambda}{x}}}{u_L + \frac{iu_L'}{1-\frac{\lambda}{x}}} e^{-2i(x-\lambda\log(2x)+\sigma_L-\frac{1}{2}L\pi)} \tag{52}$$

Thus, through the radial wave function at the region's infinity, the above phase shift will stabilize at a specific value, indicating a match between the wave function and the standard Coulomb wave function.

However, in our actual calculations, we found that it is not easy to match the standard Coulomb wave with the radial wave function for specific values of ω and l. Often, slow convergence issues arise, especially for $\omega \gg l$ and $\omega \ll 1$. Through our analysis, we found that the gravitational potential near the black hole creates a potential barrier similar to a "peak," which increases with increasing l. This can be deduced from the Teukolsky equation analysis. For large l, the gravitational potential near the event horizon is much greater than the energy of the scalar wave, resulting in phenomena similar to elastic scattering. Hence, the calculation of phase shifts for scattered and incident waves matches slowly. Additionally, for high ω cases, where the energy of the scalar wave is much greater than the gravitational potential, the response of the scalar wave to the potential barrier is not significant enough, resulting in slow convergence as well.

For this situation, we refer to the method used by Professor Zhang Hong to handle the scattering of wormhole potential barriers, as well as a series of works dealing with similar problems . We will use the WKB approximation method, which greatly improves the convergence speed and yields phase shift errors within an acceptable range. Therefore, this is the work we need to accomplish next.

4. CONCLUSION

In this work, we aimed to investigate the scattering of waves under long-range potentials (such as gravitational and Coulombic potentials). However, due to the divergence issue of the traditional partial wave method under long-range potentials, we proposed a new methodological framework. Instead of assuming that the incident wave is a plane wave

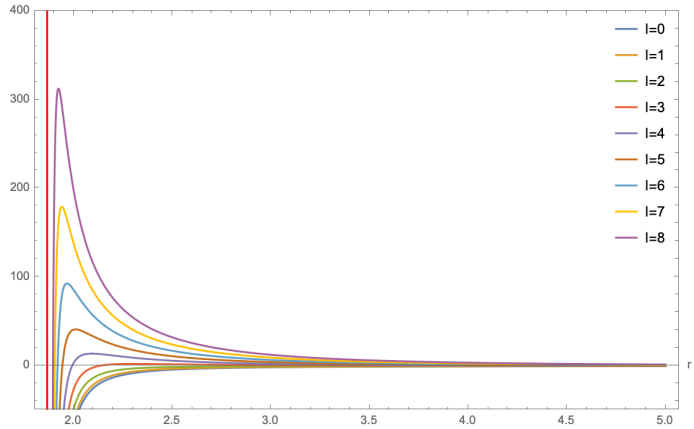

Figure 4. For different l, schematic diagrams of the black hole potential barrier are shown below. The red line on the left represents the event horizon radius.

propagating to the scattering source at infinity, we considered the joint action of long-range forces between the wave source and the scattering source. We analyzed the entire wave propagation process. To simplify the problem, we divided the system into different regions: the long-range region, the approximation region (dominated by classical Newtonian potential), and the local region (with significant general relativistic effects). Different approximations were made in different regions, which also helped to better understand the physical information in each region. For the approximation region, we used spheroidal coordinates to derive the spheroidal equation describing the wave function. Due to the difficulty in analytically solving the equation, we extracted information about scattering by expressing the wave function in series. For the local region, we considered the general relativistic effects produced by Kerr black holes and derived the spin-weighted spheroidal functions that determine the angular direction and the Teukolsky equation that describes the propagation behavior of waves near Kerr black holes in Newman-Penrose formalism. Subsequently, to consider the scattering behavior of waves in the local region, we approximately calculated the phase shift information produced by scattering in the local region. However, we encountered the problem of slow matching speed in actual calculations. Through analysis, we found that the potential barrier near the black hole responds differently to waves of different energies, leading to slow matching speed in certain specific ranges. Therefore, we referred to the works of Hong Zhang's work , and used the WKB method to approximate the phase shift, which will have ideal computational speed and accuracy. At the same time, starting from scalar waves, our theoretical framework can also be extended to the scattering calculation of electromagnetic waves and gravitational waves, providing foundational theoretical support for astronomical observations.

Overall, the article aims to propose a new approach and computational framework to solve long-standing problems in wave scattering theory, which has significant theoretical significance and application prospects.